Module Description

EI10008: Machine Learning and Data Science

Department of Computer Engineering

Module level: Bachelor	**Language:** English	**Module duration:** one semester	**Occurrence:** winter semester
Credits*: 5	**Total number of hours:** 150	**Self-study hours:** 105	**Contact hours:** 45

* The number of credits can vary depending on the corresponding SPO version. The valid number is always indicated on the Transcript of Records or the Performance Record.

Description of achievement and assessment methods:
The module examination is based on a written exam (60 minutes) which contains questions to assess the students' knowledge about the technical systems and their theoretical background, short mathematical problems to assess the students' mastering of the practiced mathematical concepts, and conceptual questions (e.g., about design principles or fundamental limitations) to assess the further intended learning outcomes. Up to 20% of the examination can be conducted in the form of multiple choice questions.

Possibility of re-taking:
In the next semester: No
At the end of the
semester: Yes

(Recommended) requirements:
Basic knowledge of linear algebra and statistics.

The following module should be successfully completed prior to participation:
MA9711 Mathematics in Natural and Economic Science 1.

The following modules are recommended to be attended in parallel (if not already attended):
MA9712 Statistics for BWL,
EI10007 Principles of Information Engineering.

Contents:
- Paradigms in data preparation
- Curse of dimensionality
- Supervised and unsupervised learning
- Principal component analysis
- Clustering algorithms
- Classification methods
- Regression methods

- Learning and generalization
- Deep neural networks

Study goals:
After attending the module, the students:
- can describe the main principles of how to retrieve information out of a huge amount of data and to reduce its dimension in an intelligent way
- are familiar with fundamental design principles of such methods and understand why existing algorithms are designed the way they are
- understand the principles of widespread machine learning techniques in general and have an overview of the underlying mathematical principles
- understand the fundamentals of deep convolutional networks and how to apply them on practical applications

Teaching and learning methods:
The module is designed for non-engineering students (in particular students in Management and Technology) who aim at understanding the fundamental principles and concepts of modern information transmission and processing. It consists of lectures, exercise courses, and self-study. In the lectures, both theoretical backgrounds and technical implementations are introduced and discussed. Mathematical concepts are introduced and explained as far as it is necessary for understanding the technical systems. The relevance of each of the considered topics is motivated by, e.g., press articles, teaser questions, or examples from daily life, and an additional reflection at the end of each topic of what has been learned aims at conveying the engineering perspective on the considered problems and systems. New concepts are presented in a teacher-centered style and discussed in an interactive manner. The aim of the exercise courses is to repeatedly practice the application of the mathematical concepts as well as the ability to answer conceptual questions about the subject. The exercise courses are held in a student-centered way, and problem sheets are provided. Throughout the semester, short reading assignments may be given to the students, e.g., as an introduction to a new topic. In addition, the students are expected to recapitulate the lecture contents and to individually practice the exercises.

Media formats:
- Slide Presentations
- Blackboard (e.g., for mathematical details)
- Supporting documents (e.g., news articles, scientific publications) as downloads (reading assignments)
- Problem sheets as downloads

Literature:
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning, MIT Press 2016
- Sergios Theodoridis. Machine Learning, A Bayesian and Optimization Perspective, Elsevier 2015
- Kevin P. Murphy. Machine Learning, A Probabilistic Perspective, MIT Press 2012
- Richard O. Duda, Peter E. Hart, and David G. Stork. Pattern Classification, John Wiley 2001
- T. Hastie, R. Tibshirani, J. Friedman: The Elements of Statistical Learning, Springer 2009.
- Charu C. Aggarwal: Data Mining, Springer 2015

Responsible for the module:
Utschick, Wolfgang; Prof. Dr.-Ing.: utschick@tum.de

Courses (Type, SH) Lecturer:

0000005873 Machine Learning and Data Science (3SWS VI, WS 2024/25)
Utschick W, Kasibovic A

For further information about this module and its allocation to the curriculum see:
https://campus.tum.de/tumonline/wbModHb.wbShowMHBReadOnly?pKnotenNr=1485977

Generated on: 05.12.2024 18:11